

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-4

Faculty

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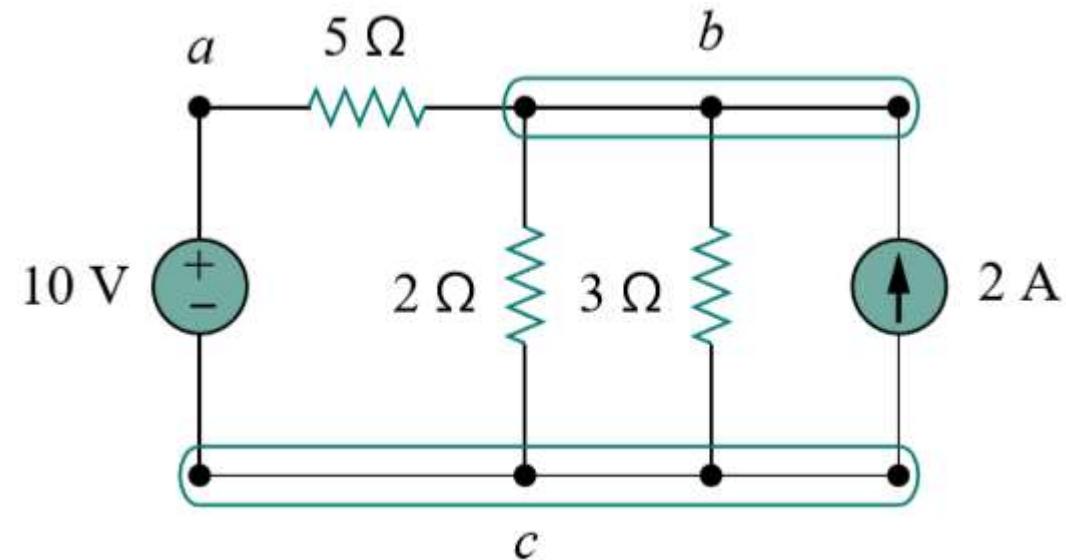
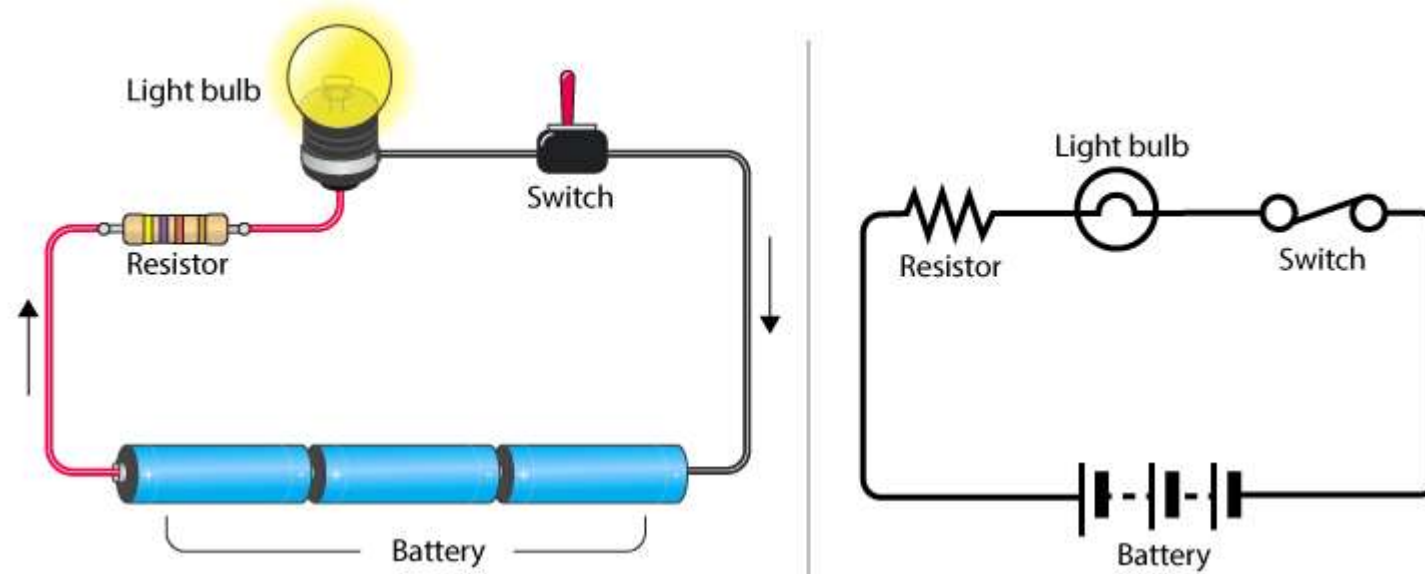
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Electric Circuit & Elements

- In Electrical engineering, we often interested in **communicating** or **transferring energy** from one point to another.
- To do this requires an interconnection of electrical devices.
- Such interconnection is referred to as an *electric circuit*, and each component of the circuit is known as an *element*.
- *An electric circuit is an interconnection of electrical elements.*

Electric Circuit



Elements/Components



Resistor



Capacitor



Inductor



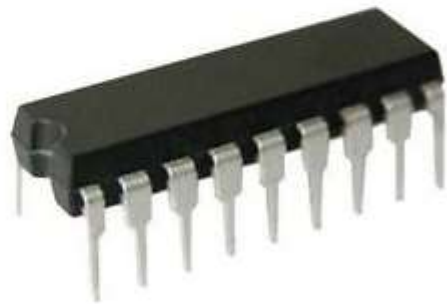
Diode



LED



Transistor



IC



Relay



Battery

Circuit Elements

There are two types of elements found in electric circuits:

Passive elements

- A passive element is not capable of generating energy
- Examples of passive elements are
 - Resistors,
 - Capacitors, and
 - Inductors.

Active elements

- An active element is capable of generating energy.
- Typical active elements include
 - generators,
 - batteries, and
 - operational amplifiers.

Circuit Elements-Passive

Resistance (R):

- It is an electrical property by which it opposes the flow of charges or current.
- Therefore the terminal in which the current enters is always has higher potential (+ve) with respect to the other terminal of the resistor.
- Unit of resistance is Ohm (Ω)
- Inverse of resistance is called as Conductance and its unit is Mho or Siemens (S).

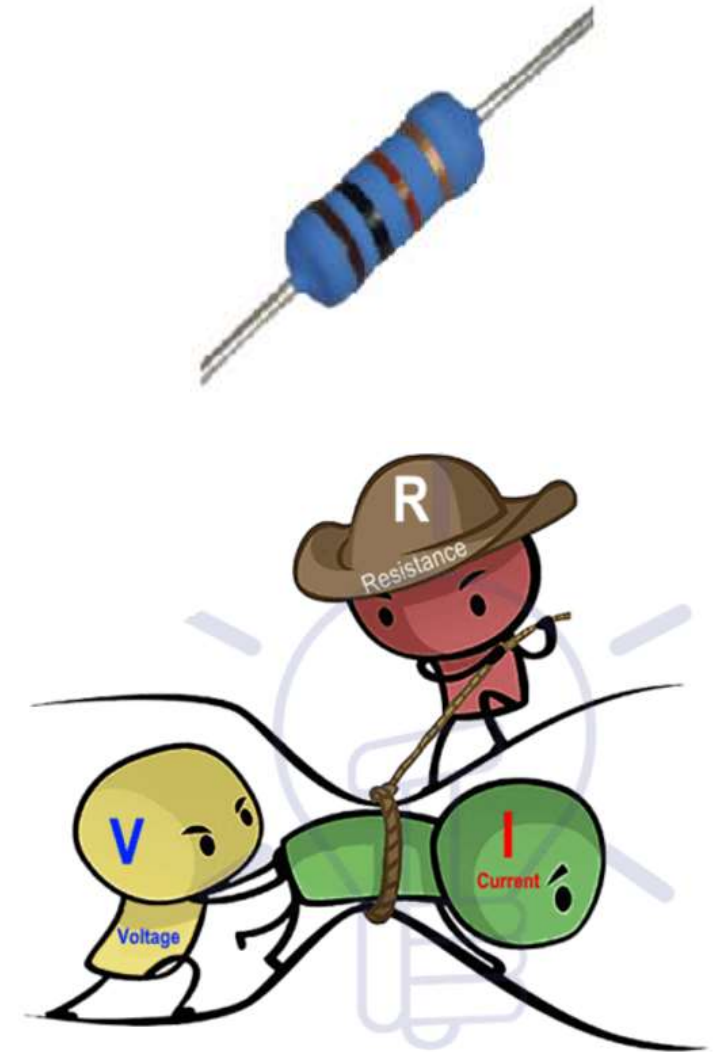
$$R = \frac{V}{I} = \frac{\rho l}{a}$$

V and I are voltage across and current through the resistance R .

l – Length in m, a – Cross sectional area in m^2

ρ – Resistivity in $\Omega - m$

- A **short circuit** is a circuit element with resistance approaching **zero**.
- An **open circuit** is a circuit element with resistance approaching **infinity**.





Carbon composition resistor



Wire wound resistors



Cermet oxide resistor



Fusible resistors



Potentiometer



Trimmer



Rheostat



SMD



Varistors



carbon film resistor



Thermistors



Photoresistors

Resistor Colour Coding

Resistors



Resistors Color Coding



4 Band

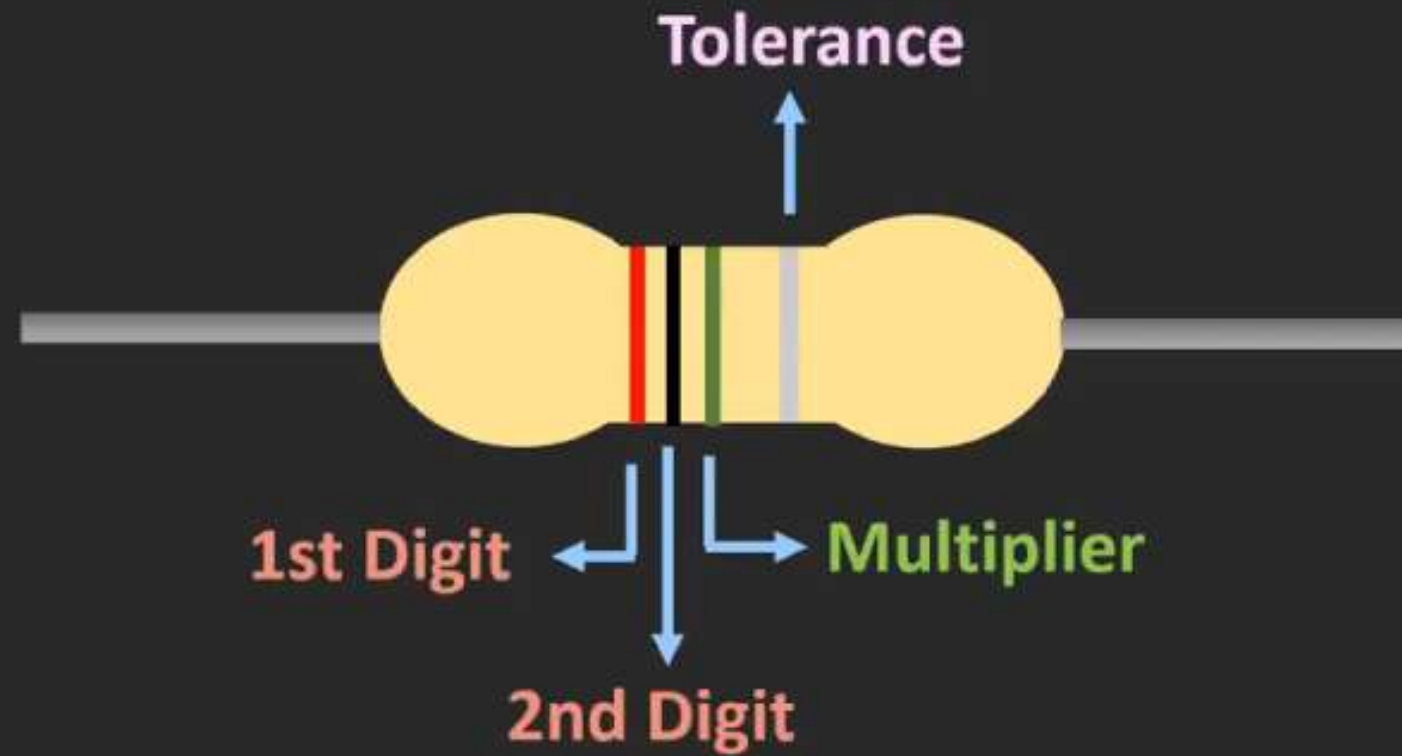


5 Band



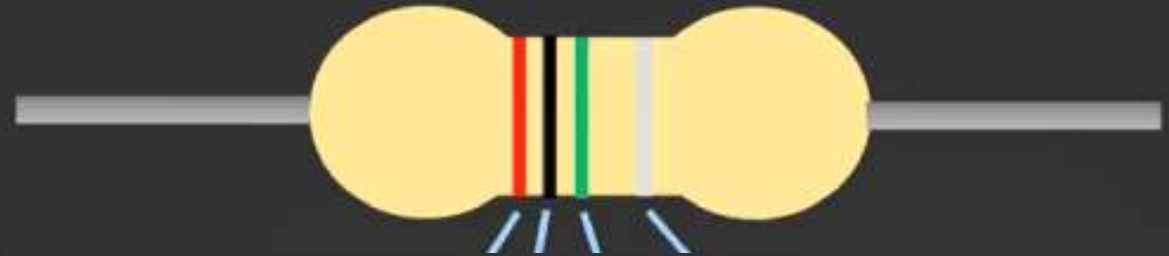
6 Band

4 Band Resistor



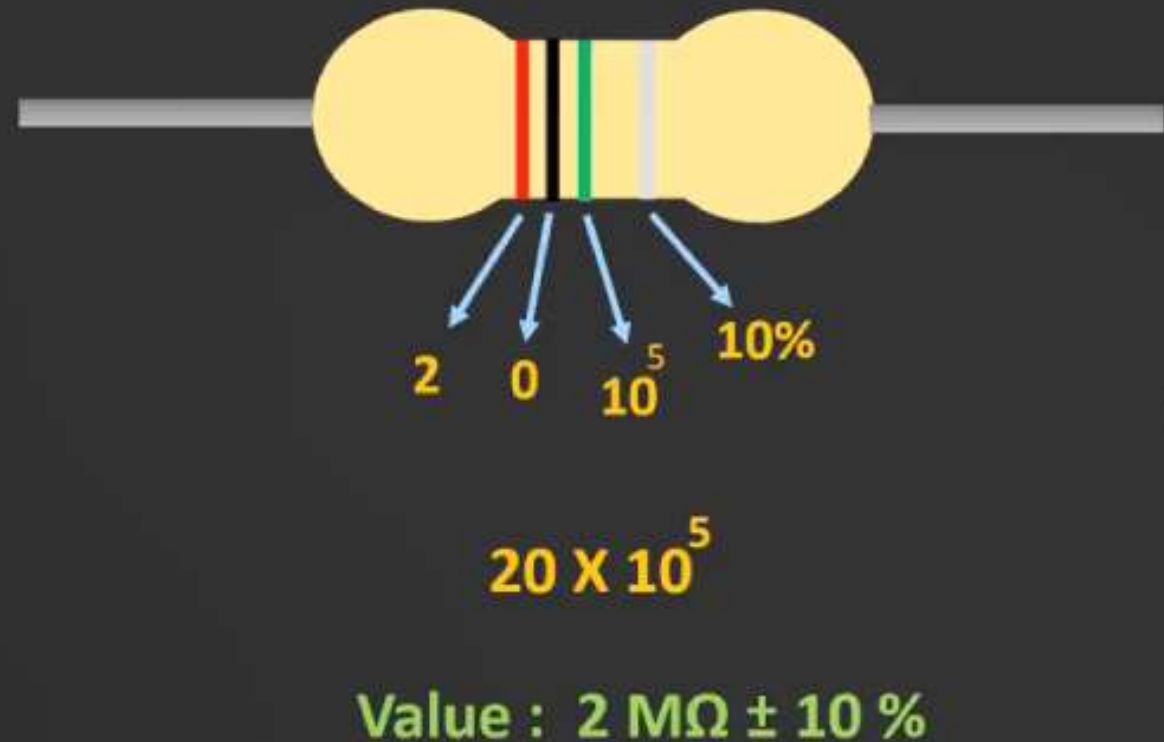
Resistor Color Coding (4 Band)

Color	1 st Digit	2 nd Digit	Multiplier	Tolerance
Black	0	0	10^0	
Brown	1	1	10^{-1}	1%
Red	2	2	10^{-2}	
Orange	3	3	10^{-1}	
Yellow	4	4	10^4	
Green	5	5	10^5	0.5%
Blue	6	6	10^{-6}	0.25%
Violate	7	7	10^7	0.1%
Grey	8	8	10^8	0.05%
White	9	9	10^9	
Gold			10^{-1}	5%
Silver			10^{-2}	10%



Resistor Color Coding (4 Band)

Color	1 st Digit	2 nd Digit	Multiplier	Tolerance
Black	0	0	10^0	
Brown	1	1	10^1	1%
Red	2	2	10^2	
Orange	3	3	10^3	
Yellow	4	4	10^4	
Green	5	5	10^5	0.5%
Blue	6	6	10^6	0.25%
Violet	7	7	10^7	0.1%
Grey	8	8	10^8	0.05%
White	9	9	10^9	
Gold			10^{-1}	5%
Silver			10^{-2}	10%



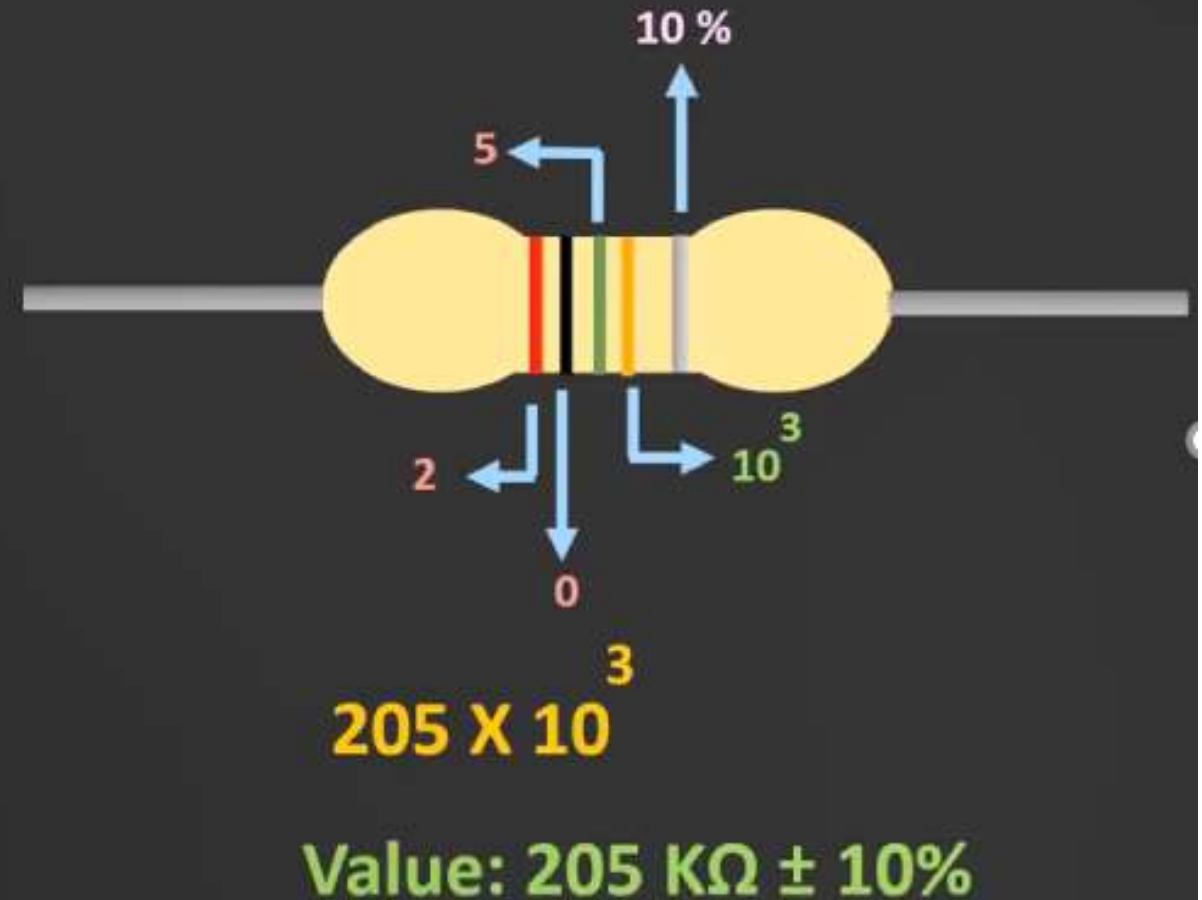
Resistor Color Coding (5 Band)

Color	1 st Digit	2 nd Digit	3 rd Digit	Multiplier	Tolerance
Black	0	0	0	10^0	
Brown	1	1	1	10^1	1%
Red	2	2	2	10^2	
Orange	3	3	3	10^3	
Yellow	4	4	4	10^4	
Green	5	5	5	10^5	0.5%
Blue	6	6	6	10^6	0.25%
Violet	7	7	7	10^7	0.1%
Grey	8	8	8	10^8	0.05%
White	9	9	9	10^9	
Gold				10^{-1}	5%
Silver				10^{-2}	10%



Resistor Color Coding (5 Band)

Color	1 st Digit	2 nd Digit	3 rd Digit	Multiplier	Tolerance
Black	0	0	0	10^0	
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Red	2	2	2	10^2	
Orange	3	3	3	10^3	
Yellow	4	4	4	10^4	
Green	5	5	5	10^5	0.5%
Blue	6	6	6	10^6	0.25%
Violet	7	7	7	10^7	0.1%
Grey	8	8	8	10^8	0.05%
White	9	9	9	10^9	
Gold				10^{-1}	5%
Silver				10^{-2}	10%





Circuit Elements-Passive

Inductance (L):

- It is an electrical property by which it opposes any change in the current flowing through it.
- It stores electrical energy in the form of electro-magnetic.
- Unit of inductance is Henry (H)
- The stored energy is $E = \frac{1}{2} L I^2$ Joules
 I is the current through the inductance L .



Types of Inductor

Molded inductor & air-wound inductor



Air wound inductor



Iron powder Toroidal inductor



Ferrite core Toroidal transformer



Adjustable air-wound inductor



Variable Inductor



Variable



Circuit Elements-Passive

Capacitance (C):

- It is an electrical property by which it opposes any change in the voltage across it.
- It stores electrical energy in the form of electro-static.
- Unit of capacitance is Farads (F)
- The stored energy is $E = \frac{1}{2}CV^2$ Joules
 V is the voltage across the capacitance C .



Capacitor types



Ceramic disk



**Monolithic
ceramic**



**Dipped silver-
mica**



Mylar



Mylar



**Solid tantalum,
polarized**
LSU 08/11/2004



**Radial aluminum
electrolytic**

Electronics 2



**Axial aluminum
electrolytic**

Tantalum



Trimmer



Aluminum



Thin Film



Ceramic

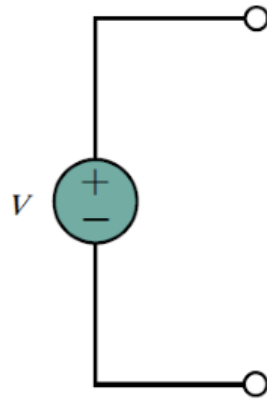


Circuit Elements-Active

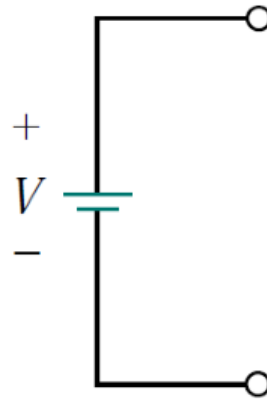
Power sources:

- The most important active elements are **voltage or current** sources that generally deliver power to the circuit connected to them.
- There are two kinds of sources:
- **Independent sources**
 - An ideal independent source is an active element that provides a specified voltage or current that is completely independent of other circuit variables.
 - Independent sources are usually designated by circular shaped symbols.
- **Dependent sources**
 - An ideal dependent (or controlled) source is an active element in which the source quantity is controlled by another voltage or current.
 - Dependent sources are usually designated by diamond-shaped symbols.

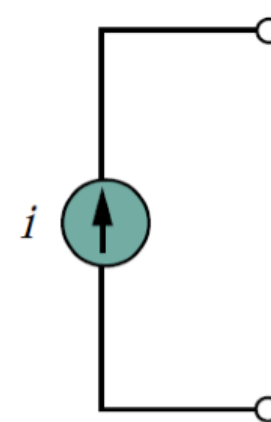
Symbols for independent sources



Voltage source which is used for constant or time-varying voltage.

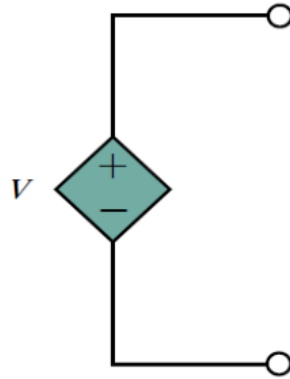


Voltage source which is used for constant voltage (dc)



Current source which is used for constant or time-varying current.

Symbols for dependent sources



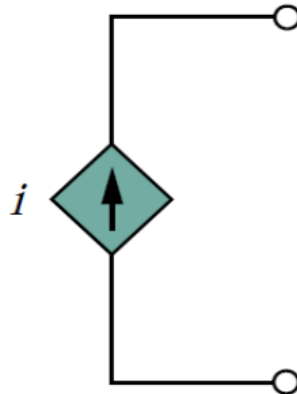
Voltage source which is depend on another current or voltage in the same circuit namely:

1. A voltage-controlled voltage source (VCVS).

$$v = x \cdot V_{AB}$$

2. A current-controlled voltage source (CCVS).

$$v = x \cdot I_{AB}$$



Current source which is depend on another current or voltage in the same circuit namely:

1. A voltage-controlled current source (VCCS).

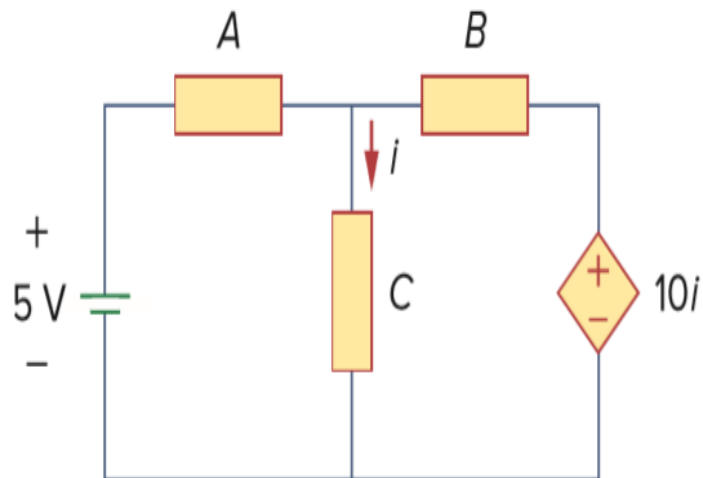
$$i = x \cdot V_{AB}$$

2. A current-controlled current source (CCCS).

$$i = x \cdot I_{AB}$$

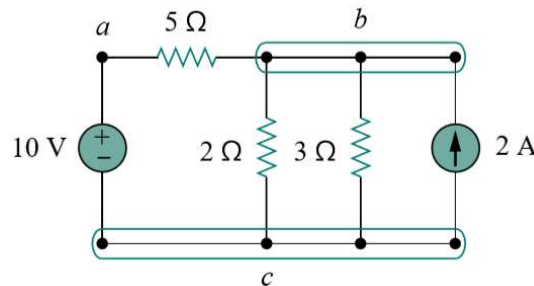
- The source quantity is controlled by another voltage or current.
 1. A voltage-controlled voltage source (VCVS).
 2. A current-controlled voltage source (CCVS).
 3. A voltage-controlled current source (VCCS).
 4. A current-controlled current source (CCCS).

Identify Different sources in below circuit



Branches, Nodes and Loops

- A **branch** represents a single element such as a voltage source or a resistor.
- A **node** is the point of connection between two or more branches.
 - Minor nodes : interconnection of only two branches and there is no current division taking place. ($I_1 = I_2$)
 - Major nodes : interconnection of more than two branches and there is a current division / accumulation taking place. ($I_1 = I_2 + I_3$)
- A **loop** is any closed path in a circuit.
- A **mesh** is also a loop but which does not contain any other loop inside.



Identify the branches, nodes, loops and meshes

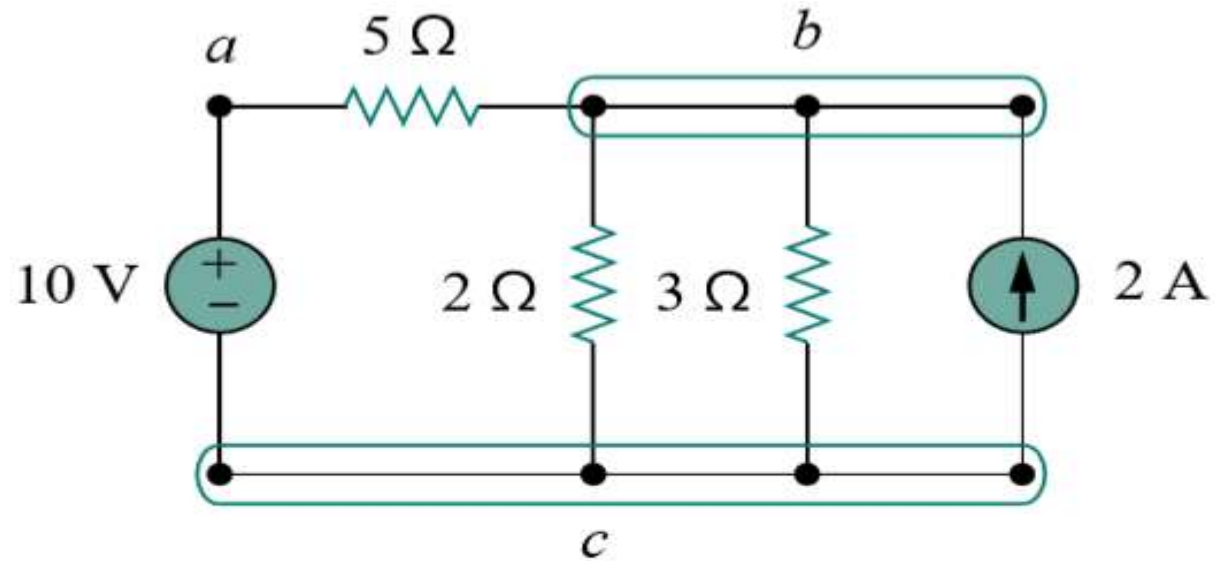
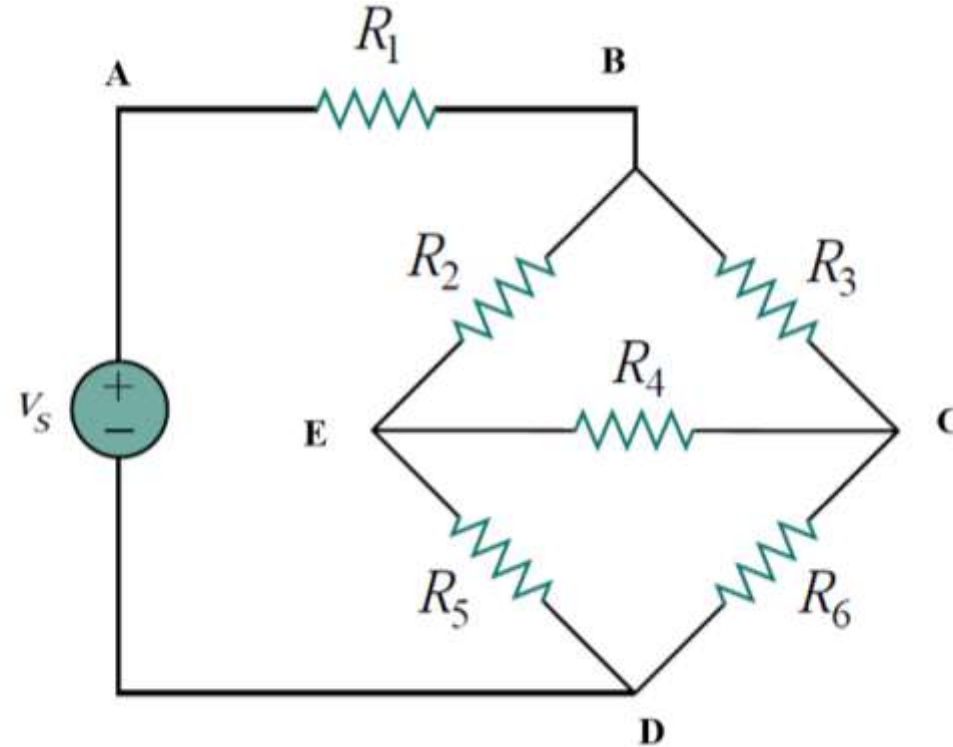


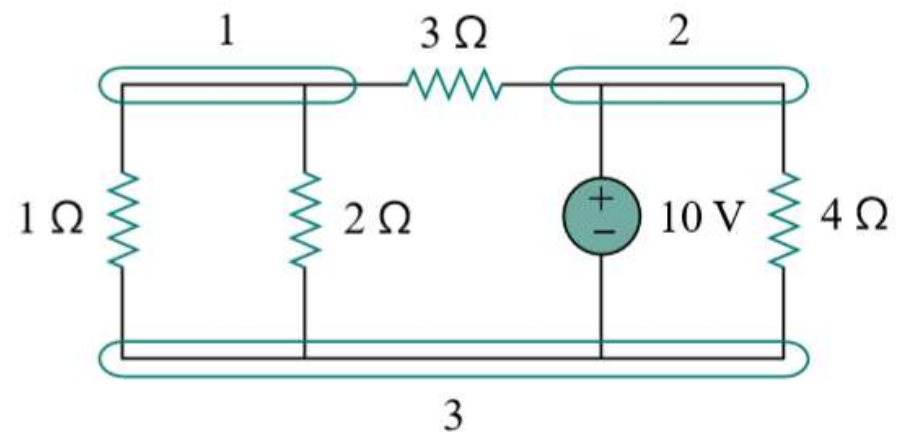
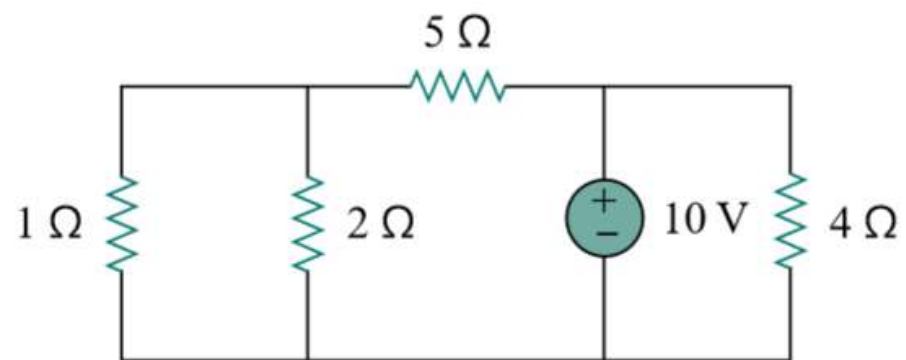
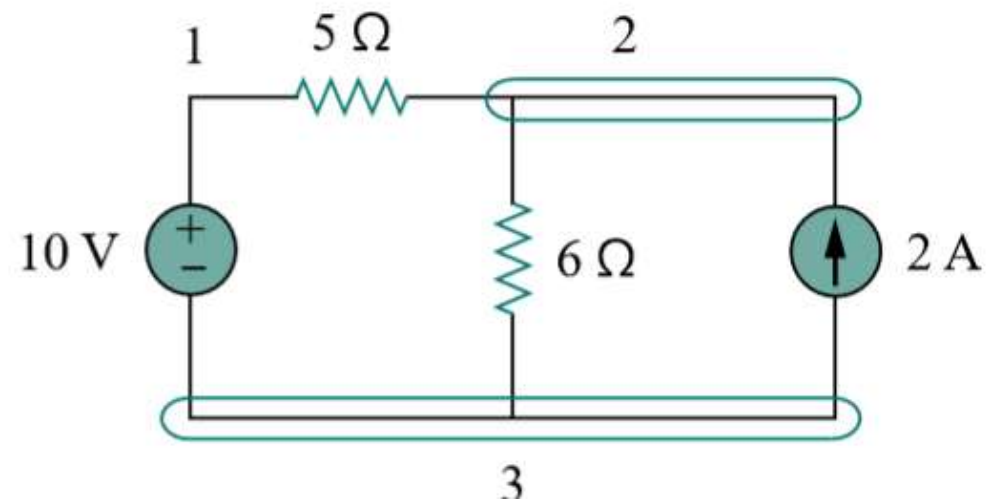
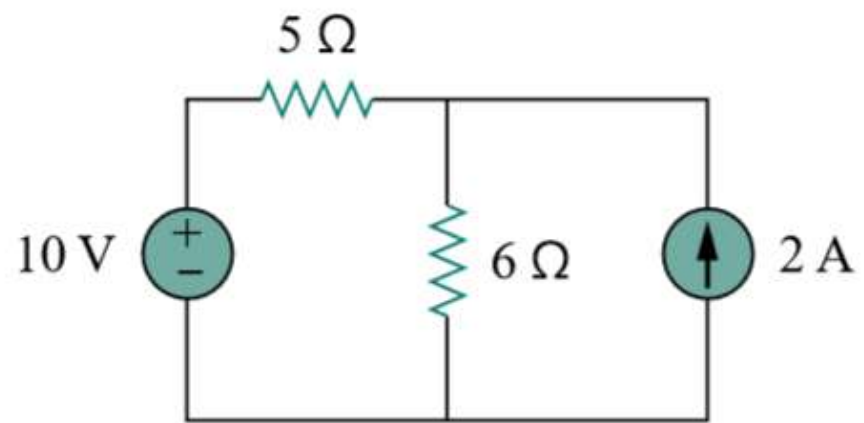
Figure Nodes, branches, and loops.

Identify the branches, nodes, loops and meshes



Branches are AB, BC, CD, DE, CE, EB and DA Nodes are A, **B**, **C**, **D** and **E**.

Loops are **ABCD**, **ABECDA**, **ABCEDA**, **BCDEB**, BCEB, CDEC and ABEDA.

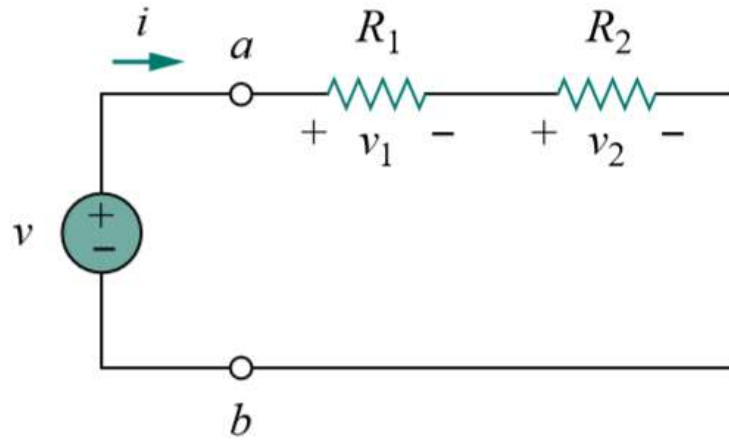


Simple circuit analysis – Voltage and Current division principle

Series & parallel

- **Series:**
 - Two or more elements are in series if they are cascaded or connected sequentially and consequently carry the same current.
- **parallel:**
 - Two or more elements are in parallel if they are connected to the same two nodes and consequently have the same voltage across them.

Series R



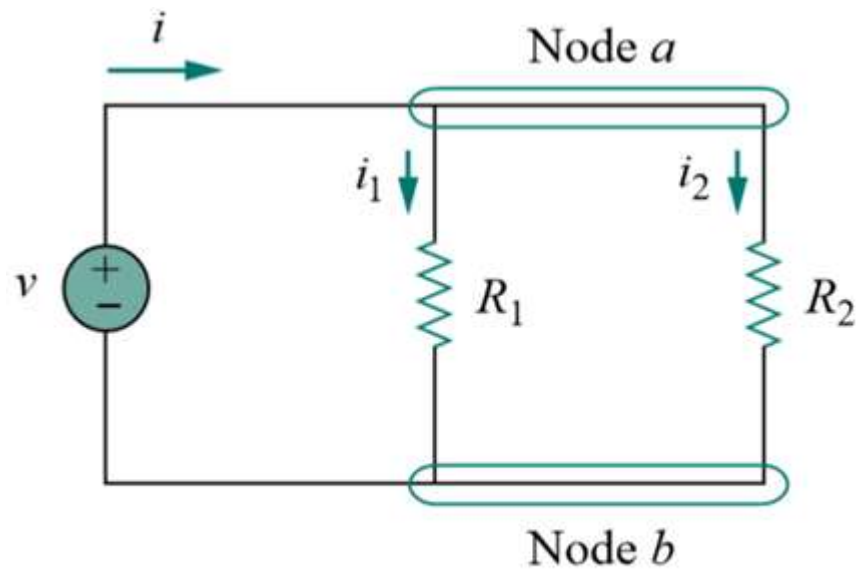
A single-loop circuit with two resistors in series.

$$R_{\text{eq}} = R_1 + R_2$$

$$R_{\text{eq}} = R_1 + R_2 + \cdots + R_N = \sum_{n=1}^N R_n$$

The **equivalent resistance** of any number of resistors connected in series is the sum of the individual resistances.

Parallel R



Two resistors in parallel.

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{\text{eq}}} = \frac{R_1 + R_2}{R_1 R_2}$$

$$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$$

Find 'R' equivalent

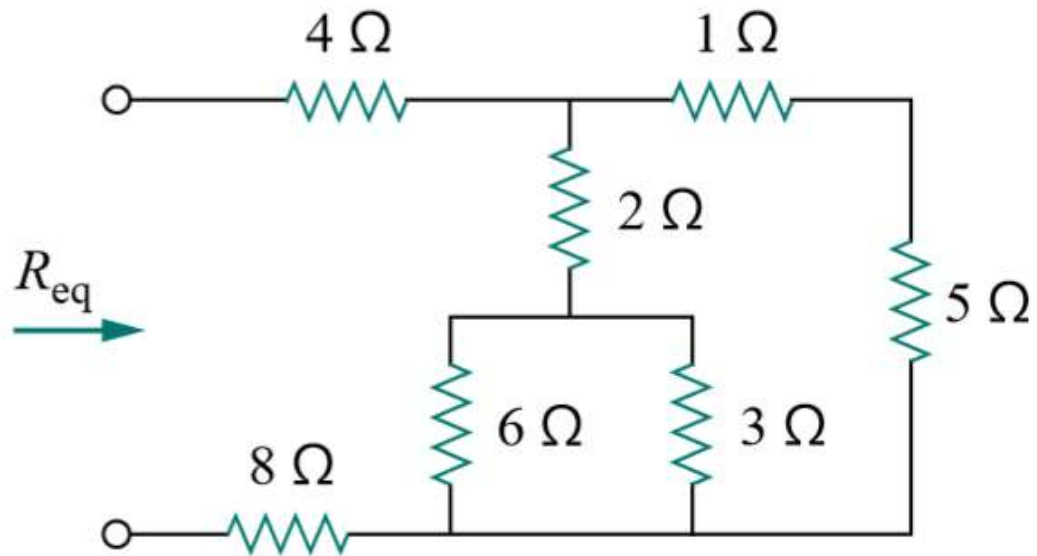


FIG:1

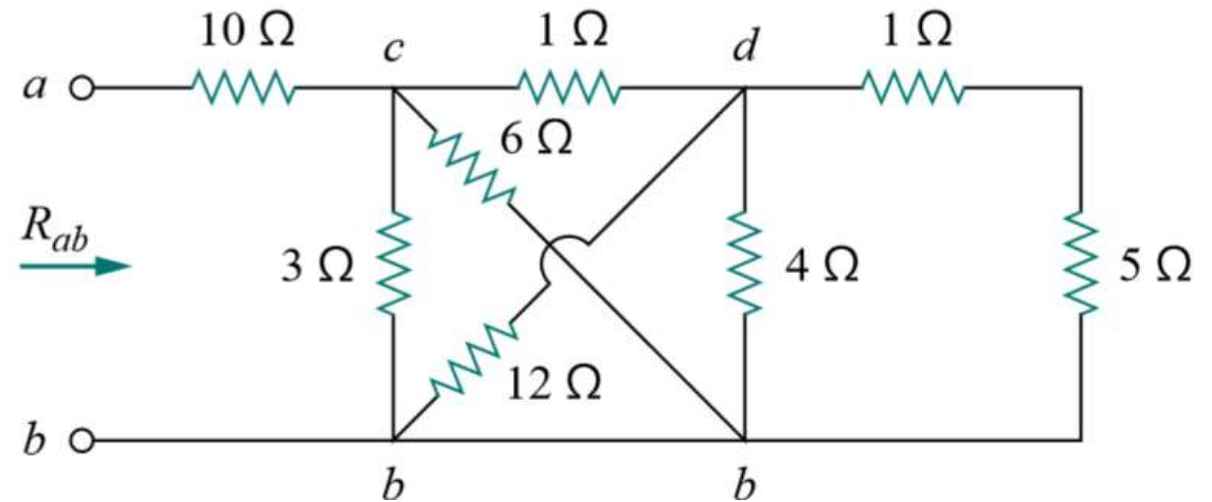
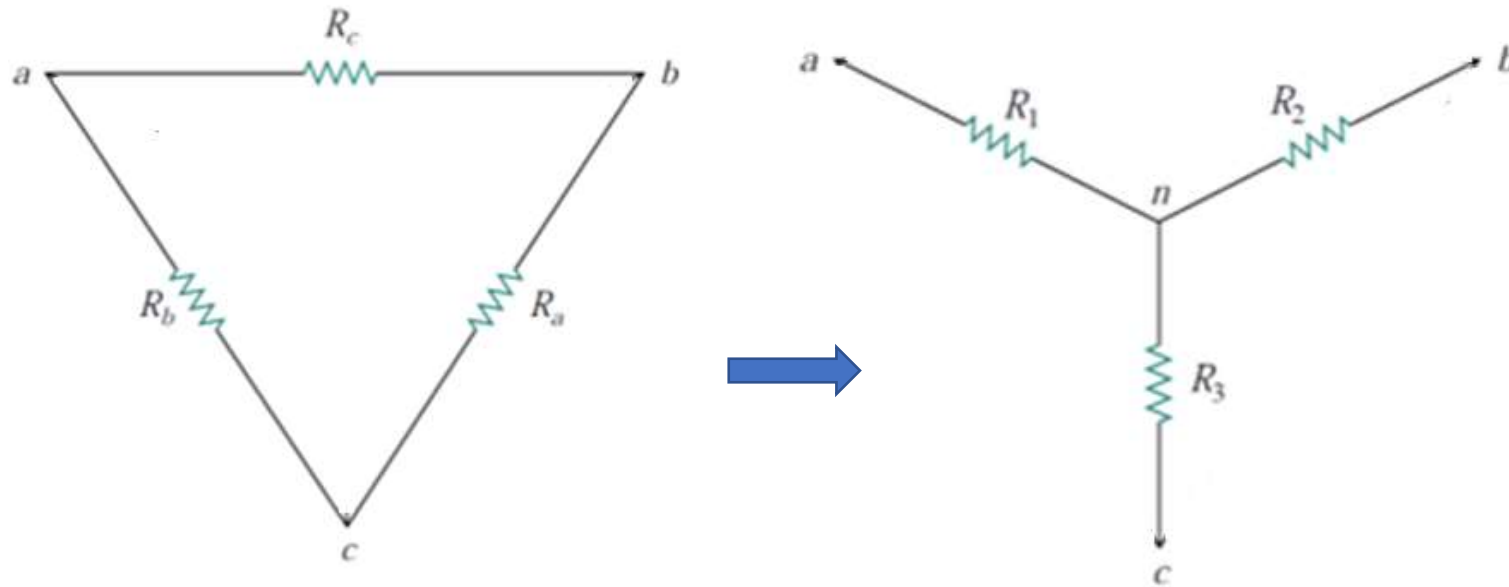


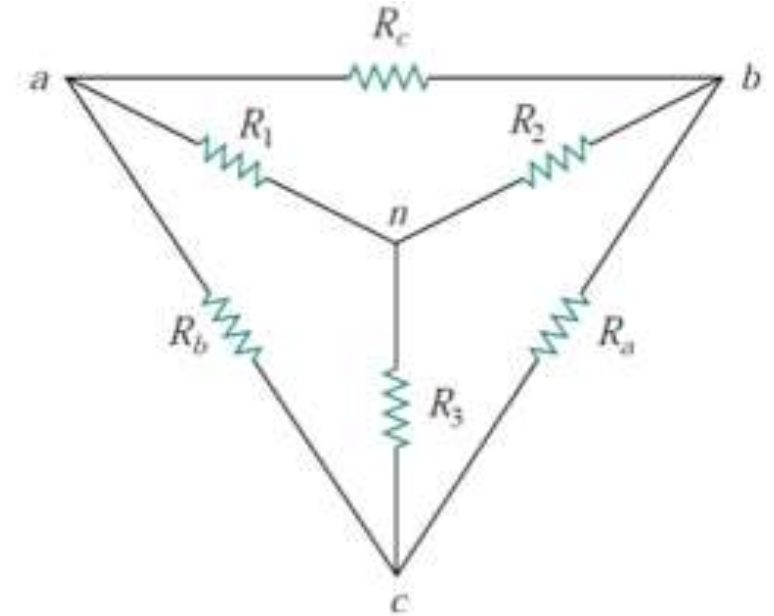
FIG:2

1.DELTA(Δ) to STAR(Y)



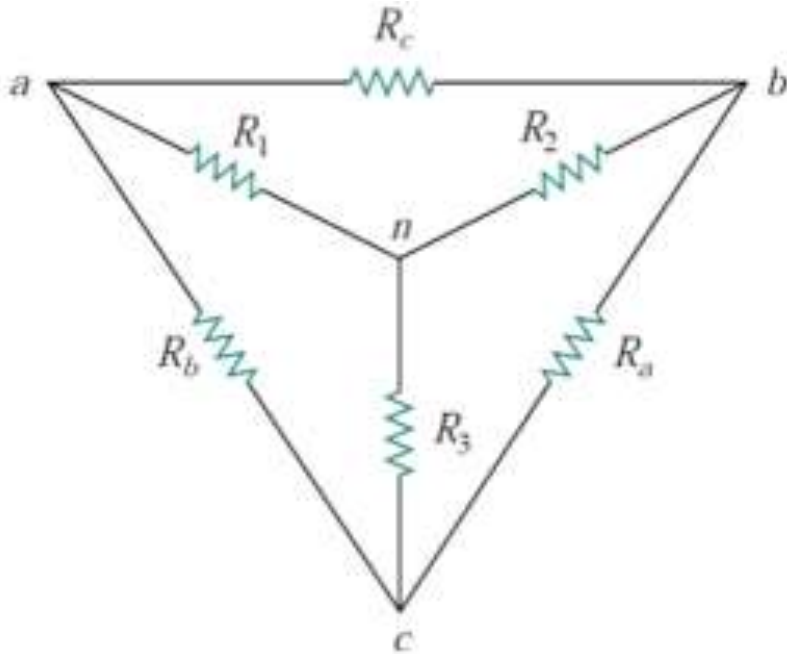
$$R_1 = \frac{(R_b R_c)}{(R_a + R_b + R_c)}$$

$$R_2 = \frac{(R_a R_c)}{(R_a + R_b + R_c)}$$



$$R_3 = \frac{(R_a R_b)}{(R_a + R_b + R_c)}$$

DELTA(Δ) to STAR(Y)



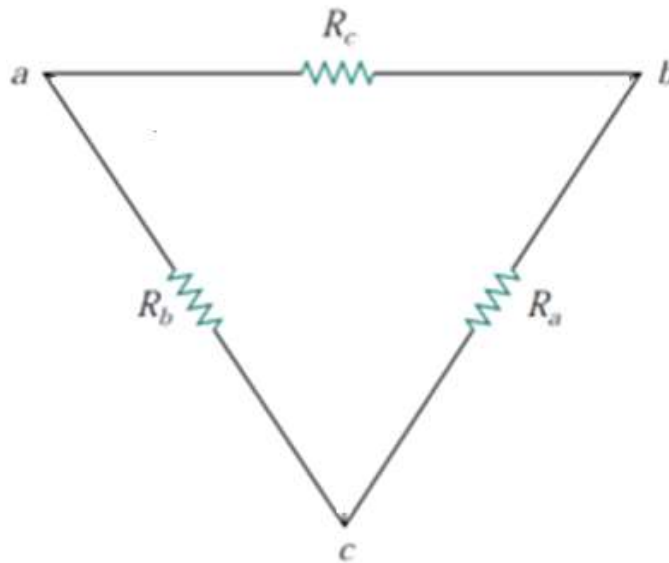
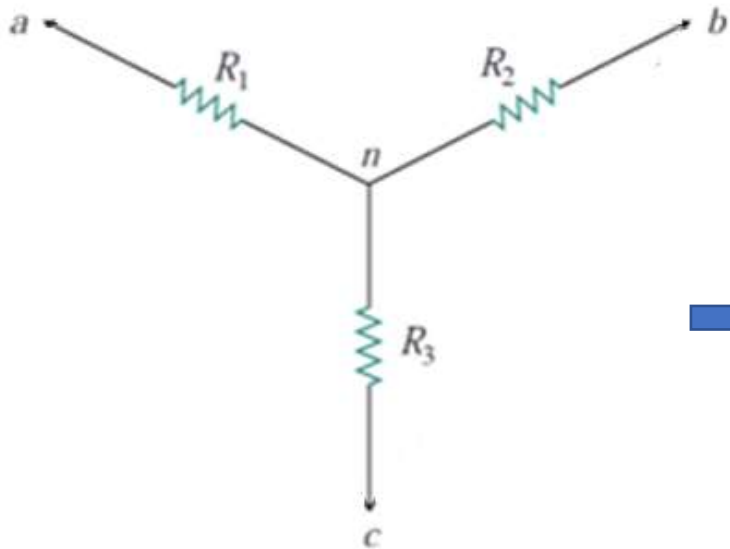
$$R_1 = \frac{(R_b R_c)}{(R_a + R_b + R_c)}$$

$$R_2 = \frac{(R_a R_c)}{(R_a + R_b + R_c)}$$

$$R_3 = \frac{(R_a R_b)}{(R_a + R_b + R_c)}$$

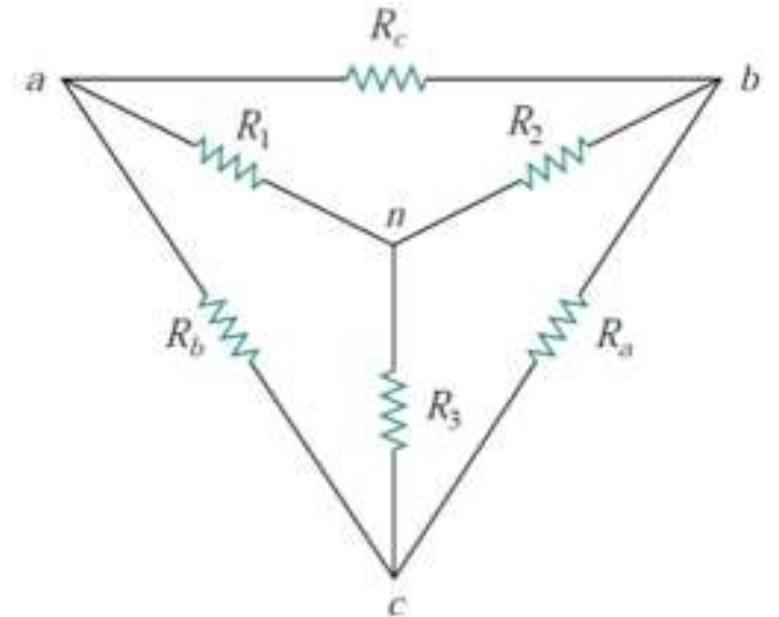
- Each resistor in the STAR network is the product of the resistors in the two adjacent DELTA branches, divided by the sum of the three DELTA resistors.

2.STAR(Y) to DELTA(Δ)



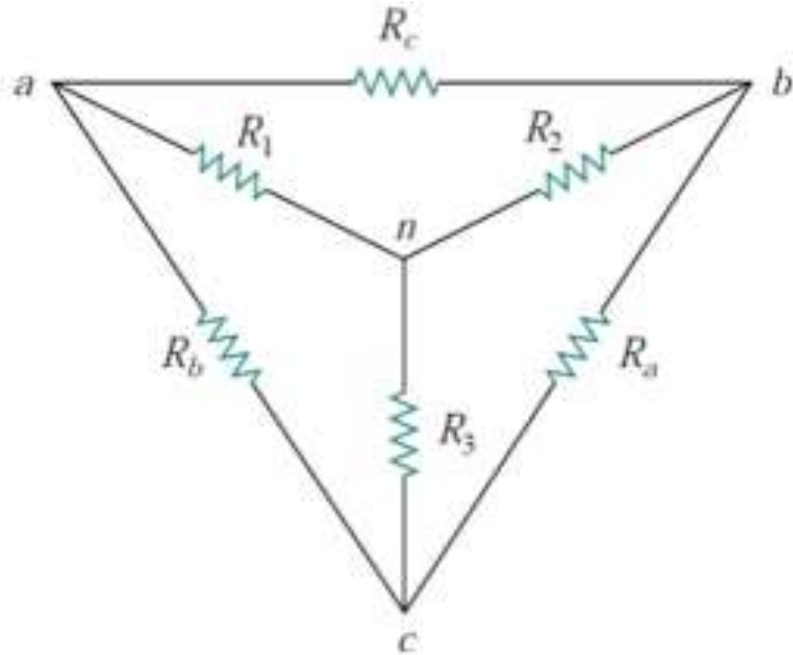
$$R_a = \frac{(R_1 R_2) + (R_2 R_3) + (R_3 R_1)}{R_1}$$

$$R_b = \frac{(R_1 R_2) + (R_2 R_3) + (R_3 R_1)}{R_2}$$



$$R_c = \frac{(R_1 R_2) + (R_2 R_3) + (R_3 R_1)}{R_3}$$

STAR(Y) to DELTA(▲)



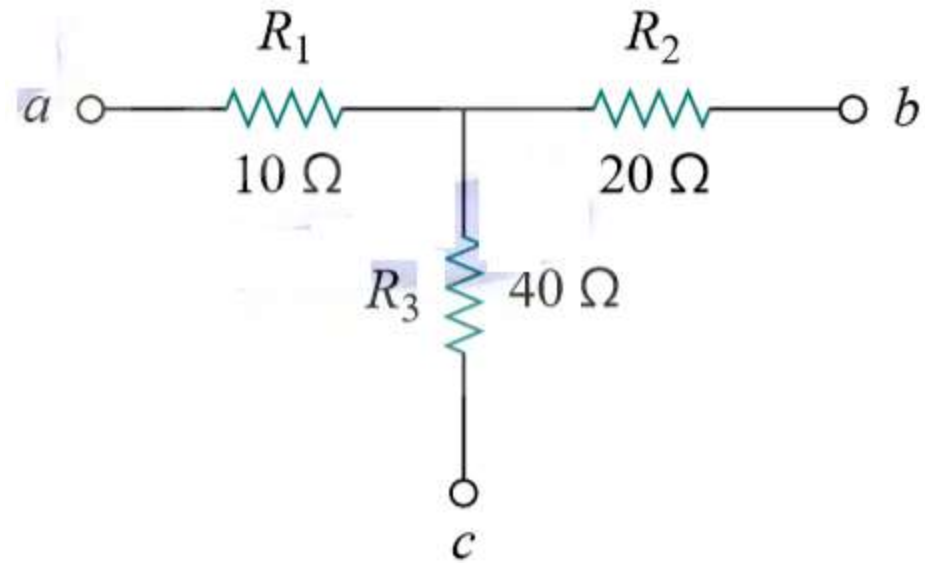
$$R_a = \frac{(R_1 R_2) + (R_2 R_3) + (R_3 R_1)}{R_1}$$

$$R_b = \frac{(R_1 R_2) + (R_2 R_3) + (R_3 R_1)}{R_2}$$

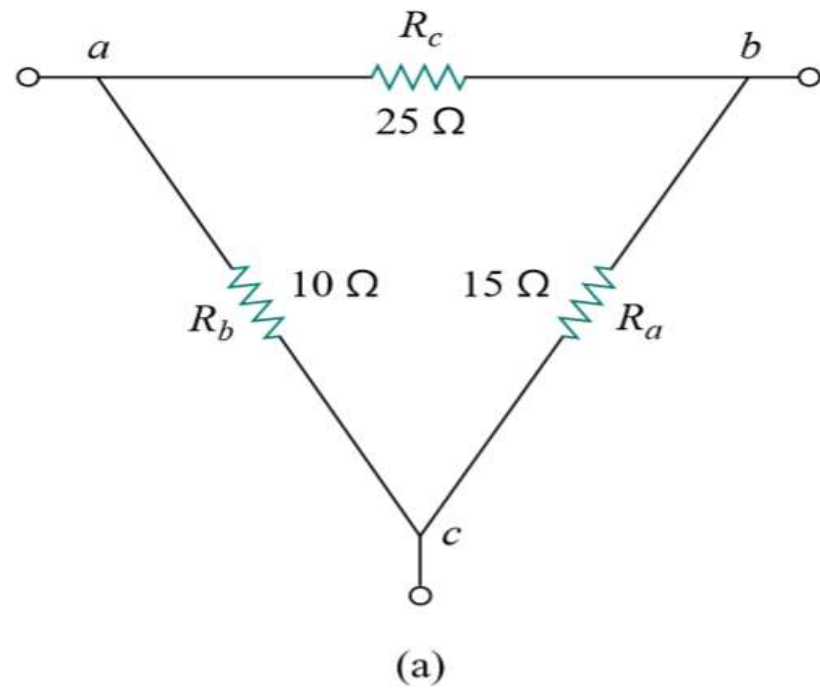
$$R_c = \frac{(R_1 R_2) + (R_2 R_3) + (R_3 R_1)}{R_3}$$

Each resistor in the Δ network is the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistor.

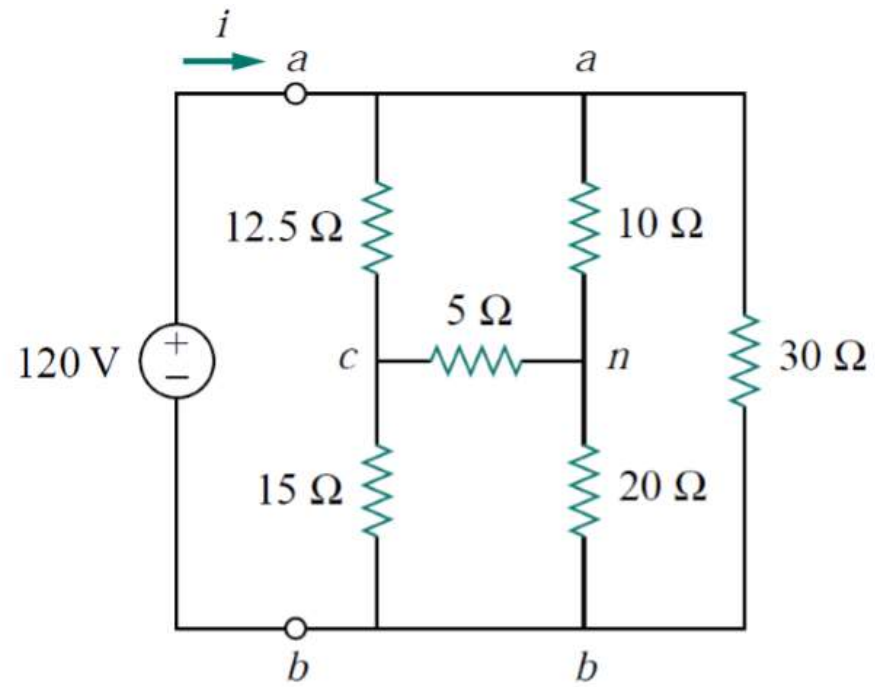
Problem-1: Convert Star to Delt Network



Problem-2: Convert Delt to Star Network



Problem-3: Find R_{ab}



Problem-4

Find the R_{ab}

